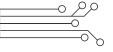




Graph Presence Is Not Graph Dependence: A Controlled Audit of Structural Utility in Multimodal Music Understanding

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ABSTRACT

We test whether a previously built multimodal music-understanding system's graph neural network actually depends on its intended graph topology, after controlling for edge density, pooling choice, and node content. Our first attempt to randomize the graph's edges found no effect. But a forensic check showed why: the graphs we started from were nearly complete (76–90% fully complete, depending on dataset), so a randomized graph mostly reconstructed the same edges under a different name. We repaired this with a sparser, pre-verified graph construction and repeated the audit. Across nine dataset, neighborhood-size, and pooling combinations, we found no statistically reliable training-time advantage from the intended topology. This is a failure to detect an advantage, not proof of equivalence. A positive control confirms our method can detect engineered topology dependence at every density we test, with one exception: max-pooling's sensitivity to a node-relabeled graph weakens as density increases, and disappears completely at the highest density tested. At evaluation time, three of eighteen comparisons survive correction for multiple comparisons, all in the sparsest regime we test (GTZAN, $k=4$), where predictions do reliably depend on the specific graph shown. A post-hoc interaction test rejects equality across the three regimes: this regime's effect reliably exceeds one of the other two, but not the third after correcting for multiple comparisons. The matched training-time comparison shows no such dependence in the same regime. Removing all edges causes a large, reliable drop in performance in every regime, so the model does depend on neighbor aggregation being present. Outside GTZAN $k=4$'s evaluation-time exception above, we did not find an equally reliable dependence on which specific neighbors are aggregated. In the deployed fusion system, pairing a caption with a different track's graph causes no statistically reliable drop in performance, whether we use one fixed mismatch or ten independently sampled mismatches evaluated across four checkpoints; this holds even after removing literal tag-vocabulary matches and applying a matched token-count control. We do not claim graphs are unhelpful for music understanding in general, or that this topology is never used – only that we did not detect reliable use of it here. We offer a controlled audit method and a checklist for separating graph presence from graph dependence in other music information retrieval systems.

KEYWORDS:

graph neural networks, structural dependence, multimodal learning, music information retrieval, controlled audit

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1 INTRODUCTION

Graph neural networks are often added to a pipeline on the assumption that relational structure between items carries information a flat feature vector does not. This assumption is easy to justify when the graph is observed independently of the features it is paired with – for example, a social network or a citation graph. It is much less clearly justified when the graph is built from the same features it will later be trained alongside: a k -nearest-neighbor graph over point-cloud coordinates, or, in our case, a similarity-thresholded graph over per-segment audio features. In this feature-induced setting, part of the graph’s structure is not a new observation about the world. It is a deterministic summary of information the model already has in its node features.

Using a graph neural network does not, by itself, tell us whether a model actually relies on the relations that motivated the architecture. Its accuracy could instead come from the node features, the pooling operator, the mere presence of a neighbor-aggregation step regardless of what it aggregates, or, in a multimodal system, a different modality entirely. A standard graph-versus-no-graph comparison usually changes several of these factors at once, so it cannot tell them apart.

This paper reports a controlled audit of one such system: a previously implemented, multimodal music-understanding pipeline whose graph branch encodes short audio segments and their similarity relations. We ask a narrower question than “is the graph useful”: does the model depend on its intended topology, once we control for edge presence, pooling, and graph density? We split this into two questions that do not imply each other. Within one graph, does the model depend on edges being present, and on which specific edges are present? Across modalities, does a fused system depend on the graph being paired with the right example, or would any plausible graph do? If a caption is paired with an unrelated track’s graph and the fused prediction barely changes, the graph branch may be present without doing any sample-specific work. Section 6 tests this directly.

Answering these questions well required repairing our own audit along the way. An early attempt at randomizing the graph’s edges found no effect. But a forensic check showed why: the construction we started from produced near-complete graphs, so most randomization attempts reconstructed almost the same graph under a different name. Our topology-identity results therefore use a separately verified, sparser construction, built to make randomization meaningful, rather than the originally deployed one. Section 5 tells this full story, in the order it happened.

Our contributions are:

- 1 A short, scoped observation, a direct consequence of the data-processing inequality, that a

feature-induced similarity graph cannot carry information about a target beyond what its node features already carry. We state explicitly what this does not cover: a graph’s non-feature-induced edges, and which model an optimizer actually finds.

- 2 A verification procedure showing that a dense, feature-induced graph can make naive edge-randomization controls meaningless, plus a repaired, pre-verified sparse construction that restores a meaningful intervention on two datasets.
- 3 An audit across nine dataset/neighborhood-size/pooling combinations, finding no statistically reliable training-time advantage from the intended topology. Alongside this, a separate, statistically tested test-time result concentrates its individually significant contrasts in one sparse regime. These are two distinct questions with two different answers, on separately-trained models of the same architecture and protocol.
- 4 A cross-modal intervention on the deployed multimodal fusion system: pairing a caption with a different track’s graph causes no statistically reliable drop in macro-F1, under a fixed mismatch and under ten independently sampled mismatch mappings evaluated hierarchically across four checkpoints. This result holds even after removing literal tag-vocabulary matches and applying a matched token-count control.
- 5 A practical checklist (Section 7), drawn from what this audit needed, and in two cases had to repair, for anyone auditing a feature-induced graph branch elsewhere.

We do not claim that checking a graph’s empirical necessity, instead of assuming it, is a new idea. Section 2 credits the papers that established this idea, and states precisely what combination of the five contributions above is, to our knowledge, missing from their comparisons.

2 RELATED WORK

2.1 GRAPH NEURAL NETWORKS IN MUSIC INFORMATION RETRIEVAL

Graph neural networks are an active modeling choice across music information retrieval: heterogeneous audio-lyric graphs for emotion recognition (Mendes da Silva et al., 2022), multi-relational graphs over listening histories and tags (Peintner et al., 2025), and content plus social-interaction graphs for auto-tagging (Huang et al., 2026). These papers ask whether adding a graph improves accuracy, a design-time question, and several use graphs that are at least partly independently observed, unlike the similarity-threshold construction we study. We ask a later-stage question instead: given a model that already uses a feature-induced graph, does

it depend on that graph's intended topology once pooling, node representation, and edge-density confounds are controlled? To our knowledge, no cited music-retrieval graph paper performs this kind of controlled, after-the-fact necessity audit.

2.2 GRAPH NECESSITY AND STRUCTURE-AGNOSTIC BASELINES

The idea that graphs can be empirically unnecessary is well established. Errica et al. (2020) is the foundational result: across more than 47,000 controlled experiments spanning five GNN models and nine benchmarks, structure-agnostic baselines are competitive on several datasets. This is the same caution that motivates our own factorial, at a far smaller scale, on a deployed multimodal system rather than standard benchmarks. Duggireddy (2026) draws a closely related, contemporaneous distinction on link-prediction benchmarks between randomization sensitivity and advantage over a content-only baseline, reporting models with one but not the other. Zhang et al. (2022) distill a plain MLP from a trained graph network's soft labels and match its accuracy without graph access at inference. Katsman et al. (2026) find that a tuned MLP baseline matches or beats graph models on five of seven standard benchmarks. Bechler-Speicher et al. (2024) report a complementary failure mode: gradient descent does not reliably learn to ignore an uninformative graph on its own. This is one reason we test this empirically rather than assume it. Franceschi et al. (2019) treat an explicit or k -nearest-neighbor graph as a learnable inductive bias rather than an independent information source. This is consistent with why a feature-induced graph might still help optimization without adding information. Duong et al. (2022) study the feature-versus-structure question directly, proposing that many real graphs connect latent node features via a learnable rule – the same "nurture" premise our $A_{\text{sim}} = g(X)$ construction (Section 3) builds on. Micheli and Tortorella (2023) find that graph-rewiring benefits are often architecture- and task-specific rather than universal, a caution we share. Maganti (2026) reports a case where explicit graph structure actively hurts a downstream model relative to a structure-free alternative, a reminder that a graph branch's cost is not always zero.

We do not claim novelty from the competitiveness of structure-agnostic baselines, or from "graph necessity" in general; that credit belongs to the papers above. What this audit adds, missing from their benchmark-level comparisons, is a pre-training density-and-completeness check that catches a degenerate randomization control before it produces a misleading null; a graph-authenticity by pooling factorial that separates a topology effect from a readout confound; topology destruction on already-trained models; a cross-sample graph-pairing intervention on a deployed multimodal system; a text-dominance

control; and a positive control that validates our own methodology.

Xu et al. (2019) characterize what a message-passing layer can distinguish among multisets of neighbors. We ask a different question: whether the structure carries information beyond the multiset it was built from. Deep Sets (Zaheer et al., 2017) and Set Transformer (Lee et al., 2019) establish the permutation-invariant function class that our set-based comparison belongs to. Dynamic Graph CNN (Wang et al., 2019), which recomputes a k -nearest-neighbor graph from the current embedding at every layer, is an architecture that our Section 3 observation applies to directly.

2.3 MULTIMODAL DEPENDENCE DIAGNOSTICS

Liang et al. (2023) separate the information available in a multimodal dataset from what a trained fusion model actually captures, which is relevant to the confound checks in Section 6. Wei et al. (2024) distinguish an intrinsically uninformative modality from a merely under-trained one, addressing the same confound. MM-SHAP (Parcalabescu and Frank, 2023) uses Shapley values to quantify how much a vision-language model relies on each modality; our approach is a direct intervention rather than an attribution method, and it targets a graph modality specifically. The congruent-versus-incongruent-pairing diagnostic originates with Elliott (2018), who find that only one of three multimodal machine translation systems is sensitive to an incongruent image at all. Li et al. (2021) revisit this and report a closely parallel finding: on the standard Multi30K benchmark, substituting an unrelated image is only marginally impactful. They attribute this null to a dataset property – translations were produced without translators seeing the images – since sensitivity does appear on a subset requiring genuine visual disambiguation. We read both papers, and our own result, as consistent instances of a broader pattern: aggregate incongruent-pairing tests often find weak sensitivity on naturally-occurring data, though this does not rule out that a more targeted diagnostic could reveal more elsewhere. We do not claim novelty from the observation that multimodal systems can ignore a modality in general; that is established by MM-SHAP and the broader unimodal-collapse literature (Du et al., 2023; Wang et al., 2020).

3 WHY GRAPH PRESENCE IS NOT GRAPH DEPENDENCE

Before describing the system we audit, it helps to separate several claims that a phrase like "the model uses its graph" can quietly bundle together. A graph branch can be present without predictions depending on whether any edges exist. A model can depend on edges being present without depending on which specific edges are

present. And in a multimodal system, a graph encoder can matter for its own sub-task while the fused output ignores whether that graph belongs to that example. These are not steps of one ladder; none implies the next. We test two directions: dependence within one graph (is neighbor-aggregation used at all, and does it matter which specific edges it aggregates over), and dependence across modalities (does the graph branch contribute to the fused prediction, and does it matter whose graph is attached to whose caption). Section 6 reports one experiment for each question.

Throughout this paper, topology identity means whether a model's output depends on the specific, labeled edge-node incidence it was trained or evaluated with, versus an equally-sized alternative. This is not the stricter graph-theoretic sense of isomorphism class: since node identity here means a specific segment with specific features, a change that leaves the isomorphism class unchanged can still count as a "different topology" in this practical sense. Section 5 specifies the exact randomization used for each intervention: pure node-relabeling for the initial density diagnosis, and a degree-preserving edge rewiring (which does not preserve isomorphism class) for the repaired-construction factorial. For either mechanism, we measure whether an intervention changed the labeled edge set using Jaccard overlap, not any claim about isomorphism.

3.1 A SHORT OBSERVATION ABOUT THE FEATURE-INDUCED PART OF THE GRAPH

Part of the graph we audit is built directly from the same per-segment audio features the model also receives as node content: two segments are connected whenever their feature similarity crosses a fixed threshold. Write $X = \{x_1, \dots, x_n\}$ for the node-feature multiset and $A_{\text{sim}} = g(X)$ for this similarity-based edge set, where g is the deterministic, symmetric threshold rule. Because A_{sim} is a fixed function of X and nothing else, it cannot tell a model anything about a target Y that X does not already tell it:

Information ceiling for the feature-induced component. For any target random variable Y , $I(Y; X, A_{\text{sim}}) = I(Y; X)$.

This follows directly from the data-processing inequality: since A_{sim} is a deterministic function of X , conditioning on it in addition to X cannot reduce the remaining uncertainty about Y (full derivation in the supplement). We keep this result simple on purpose. It is a scoped, textbook consequence of conditioning on a deterministic transform, not a new theoretical contribution. Its value is as a cheap question to ask before crediting a graph with an accuracy gain: is my adjacency a deterministic function of my node features, and have I compared against a permutation-invariant baseline on those same features first?

Three limits on what this covers matter for what follows. First, it bounds the available information, not which model an optimizer actually finds – local aggregation may still be a useful inductive bias while carrying no additional information, so Section 6 tests this empirically rather than relying on the proposition alone. Second, the graph we audit is not A_{sim} alone: the deployed construction is $A_{\text{real}} = A_{\text{sim}} \cup A_{\text{temp}}$, where A_{temp} connects consecutive segments by position, information that is not recoverable from X and so falls outside this proposition's premise. The proposition is silent on whether A_{temp} or A_{real} carries useful information, which is why the empirical audit, not this observation, is this paper's actual evidence. Third, a message-passing layer over A_{sim} and a permutation-invariant function of the same features share only the same ceiling in principle, not guaranteed accuracy in practice; the empirical comparison in Section 6 is what actually distinguishes them.

4 EVALUATED SYSTEM AND EXPERIMENTAL SETUP

We audit a previously implemented research prototype, using its frozen checkpoints. It was neither previously peer reviewed nor put into production use. Its provenance matters only for reproducibility (Appendix A.2): the graph construction, training procedure, and checkpoints are real, fixed artifacts regardless of origin, and auditing a prototype does not change the validity of an intervention performed directly on its trained weights.

4.1 TASKS AND DATASETS

The evaluated prototype includes three graph-relevant tasks across two datasets, numbered as in the original project: 10-way genre classification (GTZAN, Task 2), multi-label tag/mood classification with text-graph fusion (MusicCaps, Task 3; concatenation and cross-attention variants), and contrastive retrieval (Task 4). This audit covers only Task 2 and Task 3; Task 4 also uses a graph branch, but we do not test it here. The graph-necessity audit in Sections 5–6.3 uses two unimodal, graph-only targets, with no text branch, so the structural signal is isolated: Task 2's GTZAN classifier, and a graph-only ablation of Task 3's MusicCaps classifier (same objective and label set, text branch removed). This is why "MusicCaps $k=4$ " appears alongside the two GTZAN regimes in Table 3 and Figure 1, rather than only GTZAN. Section 6.4 then audits Task 3's full system, with the text branch present, to test cross-modal correspondence rather than graph necessity (GTZAN's known issues are noted in Section 7.3).

4.2 NODE FEATURES AND THE ORIGINAL GRAPH CONSTRUCTION

Each track is windowed into non-overlapping, fixed-length segments. Each segment's 12-dimensional chroma and 20-dimensional MFCC features concatenate into a 32-dimensional node feature vector. Edges are added in two ways: temporal-adjacency edges connect consecutive segments by position (order-encoding, not feature content), and similarity edges connect any pair whose feature cosine similarity exceeds 0.8, exactly the $A_{\text{sim}} = g(X)$ construction from Section 3. The union of these two edge sets is what Section 5 calls the reference construction: the graph this prototype was originally built around, encoded by a 2-layer GraphSAGE network (hidden/output width 64, dropout 0.4) with global mean-pool readout and a linear head, giving 13,066 trainable parameters for the Task 2 genre classifier. Whether this construction is dense enough to support a meaningful edge-randomization intervention is exactly what Section 5 investigates first, before we report any topology-dependence result.

4.3 BASELINES, METRICS, AND STATISTICAL PROTOCOL

Our main structure-free comparison replaces every GraphSAGE layer with a per-segment MLP of identical width and depth (though not identical parameter count), with no edges and no cross-node computation, followed by the same pooling and linear head, trained identically. We report macro-F1 throughout (multi-label tasks additionally report macro AUPRC). All multi-seed comparisons use small-sample t -distribution 95% confidence intervals, not the more optimistic $z = 1.96$ normal approximation; single-run comparisons are identified explicitly. The training-time and test-time factorials (Sections 6.2–6.3) use $n = 8$ seeds per condition; the positive control and the T0/T1/T2 check use $n = 3$. Most statistics in Sections 5–6 are computed by one script that reads frozen raw result files from two dated, archived rounds (two exceptions use their own separate frozen files instead); see Appendix A.2 for the breakdown.

5 STRUCTURAL DEPENDENCY AUDIT

This section describes how we verified that our topology interventions actually intervene, and how we repaired them when an initial check showed they did not. We tell it as one sequence, in the order it happened.

5.1 THE REFERENCE CONSTRUCTION IS TOO DENSE TO RANDOMIZE MEANINGFULLY

We measured the reference construction's density directly from the cached graph tensors on each dataset's test split (32-dimensional chroma+MFCC features, similarity threshold 0.8). Three quantities matter here, and

Dataset	Density	% complete	% pairs > 0.8
GTZAN ($n=150$)	97.9%	84.0%	97.8%
MusicCaps ($n=715$)	95.4%	76.4%	95.0%
MagnaTagATune ($n=500$)	98.9%	89.8%	98.9%

Table 1 Reference-construction graph density (test splits). Mean density is the average fraction of possible edges present; % complete is the fraction of graphs at or above 0.999 density; % pairs is a property of the feature space, independent of graph realization. None substitutes for the others.

Table 1 reports all three rather than collapsing them into one:

A graph this close to complete makes two standard "fake graph" controls nearly meaningless. Node-relabeling ("shuffled") reconstructs a complete graph exactly, since a complete graph is invariant to which nodes are called what. Resampling at a matched edge count ("random") is degenerate for a related reason: there are almost no non-edges left to avoid, this close to the maximum possible count. Mean Jaccard overlap between real and shuffled/randomized graphs was 0.93–0.98 across the three datasets, matching the analytically-derived chance-expected overlap to three decimal places. This is consistent with the degeneracy being explained by density alone, with no separate randomization bug needed (full per-dataset values in the supplement).

An initial factorial comparing real, shuffled, and random graphs on this construction found no difference between them. But this null result was close to guaranteed by the construction itself, before any model was even trained, regardless of whether the model would behave differently under a genuinely different graph. We do not treat that comparison as evidence either way. It is retained internally, marked invalid, and not used as evidence here.

5.2 A REPAIRED CONSTRUCTION, VERIFIED BEFORE TRAINING

We rebuilt the similarity edges using a predefined mutual k -nearest-neighbor rule instead of a fixed similarity threshold: nodes i, j are connected only if each is among the other's k nearest neighbors by feature cosine similarity, unioned with the same temporal-adjacency edges as before. We call this the audit construction. $k \in \{4, 8, 16\}$ was fixed before any downstream training or result was inspected. From here on, "shuffled" means a degree-preserving double-edge-swap (configuration-model rewiring), not the plain relabeling used in Section 5.1: relabeling never changes the isomorphism class and is too weak a test of topology-identity dependence on its own, while a genuine rewiring is a well-defined, non-degenerate intervention now that the graph is sparse.

Dataset	k	Density	Jaccard shuf./rand.	Pass?
GTZAN	4	0.277	0.127 / 0.161	Yes
GTZAN	8	0.479	0.359 / 0.323	Yes
GTZAN	16	1.000	1.000 / 1.000	No [†]
MusicCaps	4	0.405	0.247 / 0.269	Yes
MusicCaps	8	0.821	0.857 / 0.699	No
MusicCaps	16	1.000	1.000 / 1.000	No [†]

Table 2 Pre-training verification, audit construction. [†]k=16 is capped at $n-1$ on both datasets (GTZAN's 15-segment average graphs especially), reconstructing a complete graph regardless of dataset; MusicCaps k=8 fails the same 0.5 threshold on the denser end of its degree distribution.

Before training any model, we measured the same Jaccard overlap as above for every (dataset, k) combination, on the full GTZAN test split (150 graphs) and on the first 200 of MusicCaps's 715 test graphs, and checked it against a 0.5 non-degeneracy threshold fixed in advance. We applied this threshold per (dataset, k) regime; the analysis script's own coded check tests all six regimes jointly and reports FAIL as a whole, since three of the six regimes do not clear the threshold. Table 2 reports the exact values for every regime.

Training proceeded only on the three regimes that passed: GTZAN k=4, GTZAN k=8, and MusicCaps k=4. This is a structural, pre-training exclusion. The audit-construction results in Section 6 speak to these three verified regimes, not the original threshold-based construction of Section 4; both share the same design principle, but they are not the same graph. The cross-modal wrong-track intervention is the one experiment here that uses the reference construction directly, since swapping an entire graph for a different track's graph does not need sparsity to be a meaningful intervention the way relabeling or resampling does.

5.3 A POSITIVE CONTROL ON THE AUDIT METHODOLOGY ITSELF

Before trusting a null result from the repaired factorial, we checked that our own implementation can detect real topology dependence when it is actually present. We built a synthetic binary task where the label depends exclusively on graph topology: two feature groups with well-separated, class-independent prototypes (no content-only function solves it above chance), with classes differing only in whether edges connect preferentially within or across groups, at a fixed node degree of 3 (density 0.158). We reused the exact encoder and training loop from the main audit, and its random-graph resampling intervention, unchanged; the shuffled condition here uses plain node-relabeling, not the repaired audit's degree-preserving rewiring (Section 3), with 3 seeds per pooling. This density is below every repaired regime we test (0.28–0.48). On its own, this shows that our implementation detects strong topology dependence at a sparse density;

it does not show that it has matched power at the denser regimes used in the main audit.

To check sensitivity at matched density, we reran the same generator, encoder, and training loop at k=5 (density 0.263, near GTZAN k=4) and k=9 (density 0.474, near GTZAN k=8), with 3 seeds each. Under mean- and sum-pooling, the engineered dependence remains detectable at both densities, just as at the original density. Under max-pooling, however, sensitivity to the shuffled (node-relabeled) comparison degrades sharply as density increases: accuracy climbs from 0.59 (density 0.158) to 0.79 (0.263) to a fully degenerate 1.00, indistinguishable from real, at 0.474. The mechanism is unconfirmed. It may be a general max-pooling property near completeness, or plain node-relabeling leaving group-relevant structure intact for this task design even as raw edge identity changes; we did not test which explanation is correct. Random-condition accuracy under max-pooling shows no such collapse, staying clearly separated from real at every density including 0.474 (macro-F1 0.501 vs. 1.000): the degeneracy is specific to shuffled, not to max-pooling in general. The control therefore confirms sensitivity under mean- and sum-pooling, and under max-pooling against random, across the full range we audit – but not under max-pooling against shuffled at the highest density tested (GTZAN k=8, 0.479). This caveat does not transfer to the audit-construction results, since the positive control's shuffled operator differs from theirs. Full numbers are in the supplement.

6 RESULTS

6.1 POSITIVE-CONTROL VALIDATION

Real edges achieve perfect accuracy on the synthetic topology-dependent task in every pooling and seed (macro-F1 1.000 ± 0.000). Removing edges entirely gives near-chance accuracy throughout. Both destruction interventions collapse the advantage back toward chance, though not perfectly: shuffled accuracy ranges 0.50–0.59, random accuracy 0.49–0.53. This rules out a gross implementation failure to detect strong topology dependence at this sparse density. It does not, by itself, establish matched power at the denser regimes that follow, which we check directly in Section 5.

6.2 TRAINING-TIME STRUCTURAL UTILITY

Table 3 summarizes, per regime, the range of paired differences between the intended graph and each fake-graph condition, across that regime's three poolings and three comparisons (27 contrasts total, $n=8$ seeds, Appendix A.2; the full 27-cell breakdown is in the supplement). None of the 27 contrasts excludes zero after correction. This is a failure to detect a training-time advantage in any tested combination, not a demonstration of equivalence. Several intervals remain wide enough that

Dataset/ <i>k</i>	Range across 9 contrasts	Any excl. 0 (BH)?
GTZAN <i>k</i> =4	[−0.025, +0.036]	No
GTZAN <i>k</i> =8	[−0.019, +0.006]	No
MusicCaps <i>k</i> =4	[−0.008, +0.014]	No [†]

Table 3 Training-time factorial, point-estimate range per regime across its three poolings × three comparisons (*n*=8 seeds; full 27-cell breakdown with confidence intervals in the supplement). [†]Two of MusicCaps *k*=4’s nine raw *p*-values fall below 0.05 (sum-pooling, real-vs-none and real-vs-random) but neither survives Benjamini-Hochberg correction, and the first points the wrong way: real graphs perform worse than no graph in that cell.

a true effect of practical size could go undetected. But increasing the seed count from three to eight tightened every interval, without finding one anywhere.

6.3 TEST-TIME TOPOLOGY DEPENDENCE

With training-time utility undetected, we ask a distinct question of the already-trained models: does swapping edges only at evaluation time, with no retraining, change what a trained model predicts? Removing all edges at evaluation causes a large, consistent drop (0.44–0.52 macro-F1 on GTZAN, 0.05–0.07 on MusicCaps; all nine cells significant after Benjamini-Hochberg correction within their own nine-test family). This shows the classifier head depends on the neighbor-aggregation term being present. It is a severe intervention that removes the term entirely, so it does not show that any particular relational content within it is useful.

Replacing the trained topology with a shuffled or resampled alternative, by contrast, costs much less. Figure 1 shows every cell of the resulting 18-test family (three regimes, three poolings, two destruction conditions), recomputed at *n*=8 seeds. After Benjamini-Hochberg correction across the full 18-test family at *q*=0.05, three cells survive, and all three are GTZAN *k*=4: mean-pooling real-vs-random (+0.021 macro-F1, raw *p*=0.008), max-pooling real-vs-shuffled (+0.035, raw *p*=0.0001), and max-pooling real-vs-random (+0.054, raw *p*=0.001). The latter two also survive a Bonferroni bound within the same 18-test family, the stricter of the two corrections we apply. The other nine GTZAN *k*=4 and GTZAN *k*=8 cells are all positive as well, though not individually significant, and MusicCaps *k*=4’s six cells stay close to zero throughout.

We read this as the audit’s one clear positive finding, stated at the precision the tests support: on GTZAN *k*=4, three of six test-time contrasts show a statistically reliable dependence on the specific graph shown at evaluation. The other three point in the same direction but are not individually significant, and no GTZAN *k*=8 or MusicCaps *k*=4 cell is significant either. We noticed this concentration only after seeing which cells survived correction, so we treat what follows as post-hoc and exploratory. Averaging each seed’s real-minus-perturbed difference across poolings and comparing

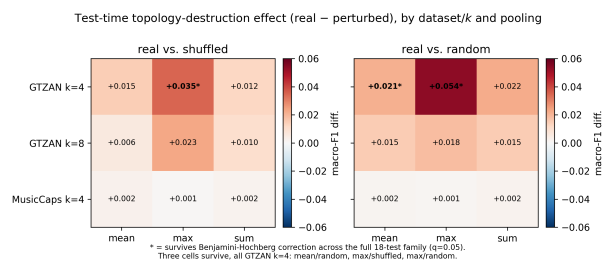


Figure 1 Test-time topology-destruction effect (real macro-F1 minus perturbed macro-F1), by dataset/*k* and pooling, shown as two separate panels so each cell’s color and label always refer to the same comparison as any marked significance. Asterisks mark the three cells, of eighteen, surviving Benjamini-Hochberg correction – all in the GTZAN *k*=4 row.

regimes with Welch’s ANOVA (robust to unequal variance) rejects equality for both destruction conditions (shuffled $F_{W(2, 10.1)}=14.1$, $p=0.001$; random $F_{W(2, 10.2)}=23.5$, $p=0.0002$). The regimes are not interchangeable, though this alone does not tell us which pairwise difference drives it. Correcting all four focal pairwise contrasts together, GTZAN *k*=4’s effect exceeds MusicCaps *k*=4’s under both conditions (Bonferroni $p=0.005$, 0.002), but its excess over GTZAN *k*=8 does not survive correction under either condition (random, the larger of the two: raw $p=0.024$, Bonferroni $p=0.094$). The defensible reading is narrower than full localization: reliably larger than MusicCaps *k*=4’s, but only directionally larger than GTZAN *k*=8’s. What does not accompany any of this is telling regardless: the matched training-time factorial (Table 3, GTZAN *k*=4) shows no reliable advantage from the real graph over shuffled or random in the same regime. One hypothesis, not an established mechanism, is that the model becomes sensitive at evaluation time to the particular graph it saw, without that being a training-time benefit. Test-time and training-time dependence answer genuinely different questions – the distinction this paper’s title refers to.

6.4 CROSS-TRACK GRAPH CORRESPONDENCE

Sections 6.2–6.3 concern the unimodal audit targets. We now ask a sharper question of the deployed, trained multimodal fusion system, using the reference construction directly (Section 5): does it matter whether the graph paired with a caption belongs to the correct track? For each test example, we compare correct pairing against a cyclic-shift mismatch (the same caption paired with a different, fixed track’s graph), with no retraining. An unchanged corpus-level macro-F1 could hide individual predictions that changed and cancelled out, so we also report example-level measures on this single checkpoint: the fraction of predictions that flip, and the mean cosine distance between the fused representation and the correct pairing’s. We separately zero out the pooled graph embedding (leaving the text branch untouched) to test

Condition	Macro-F1	Pred.-change	Cos. dist.
Correct pairing	0.6769	–	–
Graph zeroed	0.6603	1.19%	0.073
Wrong-track	0.6794	0.84%	0.088
Shuffled edges	0.6770	0.03%	0.001

Table 4 Single-checkpoint, example-level intervention effects on the deployed fusion system (shuffled edges: correct content, edges shuffled). Zeroing the graph branch changes 1.19% of predictions, wrong-track pairing a comparable 0.84%, shuffling only 0.03% – wrong-track pairing changes about as many predictions as removing the branch does, not more, against large offsetting per-example changes hiding inside an unchanged aggregate score.

Condition	Macro-F1	D (correct–wrong)	Excl. 0?
T0 – original	0.667	-0.0003 ± 0.0070	No
T1 – vocab.-sanitized	0.531	$+0.0098 \pm 0.0113$	No
T2 – matched mask	0.625	$+0.0005 \pm 0.0100$	No

Table 5 Wrong-track and caption-sanitization results ($n=3$ seeds), recomputed directly from the raw per-seed file.

whether the graph branch contributes to the fused prediction at all – the more basic question underlying the wrong-track test.

Across three freshly-trained checkpoints, correct pairing showed no statistically reliable macro-F1 advantage over this one fixed mismatch mapping ($D = -0.0003 \pm 0.0070$, 95% t -interval, $n=3$), holding the mapping fixed and varying the training seed. We also checked the reverse: fixed checkpoint, 10 independently-sampled mappings, no retraining. On the single deployed checkpoint alone, this gave a small, nominally significant disadvantage ($D = -0.0026 \pm 0.0015$, $n=10$, raw $p=0.004$). But treating those 10 mapping-level values as independent is pseudoreplication: they share one checkpoint, so the test cannot separate mapping variability from checkpoint idiosyncrasy. We therefore trained three further checkpoints (T0, seeds 0–2), reran the identical 10 mappings against each, and averaged within each checkpoint before computing a t -interval across the four checkpoint-level means. This is a crossed design (the same 10 mappings reused for every checkpoint, not independently resampled per checkpoint), so the interval describes these four checkpoints against this fixed mapping set, not checkpoints or mappings in general. Within that scope, the estimate does not exclude zero ($D = -0.0005 \pm 0.0053$, $n=4$, $p=0.79$). This is more informative than the superseded single-checkpoint value, but it is not a settled equivalence claim; both values are archived in the supplement.

A specific alternative explanation is that captions alone are so predictive that no graph condition could move the result. We retrained the fusion model under three caption conditions, three seeds each: original captions (T0); captions with every phrase matching the project’s 50-tag vocabulary removed (T1); and a matched control that removes the same number of tokens at

random rather than by semantic target (T2). Sanitization worked as intended: macro-F1 with correct pairing dropped from 0.667 to 0.531 under T1, with T2 in between. Table 5 shows that correct pairing produced no statistically reliable advantage under any condition. T1’s interval permits an effect up to roughly $+0.021$, larger than T2’s, a directional hint at three seeds. T1 removes literal matches to the project’s own tag vocabulary, so it weighs against a narrow direct-label-language shortcut, not against text dominance in general.

6.5 A FAIR ATTEMPT TO FIND A GENUINELY EXTERNAL RELATION

Perhaps our construction never delivers a genuinely external relation, one that is not recomputable from node features. Segment order is such a relation in principle, but the deployed temporal edges are symmetric, so a sequence and its time-reversal are graph-isomorphic under this encoder – order was never actually recoverable. We fixed this with directed, forward-only edges and retested the GTZAN factorial ($n=3$ seeds): the paired difference against edge-free does not exclude zero at any pooling, with all $p > 0.78$ (full numbers in the supplement). This is a null result at very low power, not a demonstrated absence of an effect. We report it as a fair test that did not pay off, not as evidence that no external relation could help; genre may simply be carried more by overall texture than by segment order.

7 DISCUSSION

7.1 WHAT THE EVALUATED SYSTEM APPEARS TO USE

Putting the results together: the classifier head depends on neighbor-aggregation’s presence in every regime. At training time, it does not detectably depend on which topology it aggregates over, in any of the nine regime/pooling combinations (three dataset/ k regimes $\times$ three poolings). At evaluation time, GTZAN $k=4$ specifically – the sparsest audit construction we test – individually shows a reliable dependence on the particular graph shown, and a post-hoc test finds its pooling-averaged effect reliably exceeds MusicCaps $k=4$ ’s, with only a directional, not significant, excess over GTZAN $k=8$ ’s. The fused system shows no reliable use of whether a caption is paired with its own track’s graph. None of these results implies any of the others; that is the point of separating them in Section 3. GTZAN $k=4$ illustrates why: a model can be sensitive to swapping its evaluation-time graph without that showing up as a training-time benefit, and without an equivalently-trained model on a different construction or dataset behaving the same way.

7.2 A STRUCTURAL-DEPENDENCY AUDIT CHECKLIST

Drawn from what this audit needed, and in two cases had to repair:

- 1 Measure density and completeness, and verify a randomization actually changes the graph, before trusting any randomization control.
- 2 Separate graph authenticity from pooling and readout using a factorial design; test edge presence and topology identity separately.
- 3 Validate the methodology with a positive control before trusting a null.
- 4 In multimodal systems, test cross-sample correspondence directly.
- 5 Control for a dominant modality precisely: matching vocabulary is not the same as removing all semantic content.
- 6 Correct for multiple comparisons within a defined family; check a stricter correction.

7.3 LIMITATIONS

Generalization. This audit concerns one system and one project's dataset family. The results do not generalize to all music graphs, all construction schemes, or domains beyond music. The cross-track result (Section 6.4) concerns one fusion architecture (cross-attention) at one fixed GNN width; the system also implements a simpler concatenation fusion with its own trained checkpoint. Whether the wrong-track null or the GTZAN $k=4$ finding replicate under a different fusion mechanism or GNN capacity remains untested, though a concatenation checkpoint already exists for the former. The system's contrastive-retrieval task (Section 4) also has a trained checkpoint but uses a different loss and evaluation metric (ranking, not classification); auditing it would need its own statistical framework rather than reusing this paper's, so we leave it untested here too.

Richer relation sources, GTZAN's known issues, and the pilot's narrow scope. Symbolic or human-annotated relational signals (chord progressions, beat grids, sections) remain untested. Our one external-relation pilot (Section 6.5) covers one relation, one architecture, and one task, so its null result does not rule out that some external signal could help. Gradient starvation (Pezeshki et al., 2021) and a mean-aggregation expressivity limit are two further plausible, untested mechanisms. GTZAN has documented repetitions, artist repetition, and mislabeling (Sturm, 2013); we use the standard split without re-auditing it. Since our one positive finding is specific to GTZAN $k=4$, we have not ruled out the possibility that near-duplicate tracks inflate this evaluation-time sensitivity estimate.

Self-audit and researcher degrees of freedom. The same group that built this prototype also designed, ran,

and interpreted every intervention reported here, including which comparisons to run and how to repair a broken control mid-study. There was no preregistration, external replication, or blinded analysis. We mitigate this the only way available after the fact: by documenting every superseded analysis, frozen input file, and post-hoc correction, rather than reporting only final numbers (Appendix A.2). But transparency about process is not independence, and a reader should weigh every result here accordingly.

Small samples and an unpreregistered correction. Confidence intervals throughout are wide enough that true effects of 0.01–0.03 macro-F1 would often go undetected. This is why we do not claim exact equivalence anywhere. The 18-test and 9-test family boundaries were fixed before inspecting which cells survive, but they were not filed as a preregistered plan in advance of data collection.

Two construction-scope caveats. First, the positive control's max-pooling sensitivity degrades with density, becoming fully degenerate at the highest density we test (Section 5), so a max-pooling null at that density is underpowered, not confirmed. Second, $k=16$ on both datasets and MusicCaps $k=8$ failed pre-training verification and were excluded, narrowing the tested density range to 0.28–0.48. Every training-time and test-time topology result concerns this audit construction, not the originally deployed reference construction directly, except the cross-track result (Section 6.4), which uses the reference construction as deployed.

8 CONCLUSION

We set out to test whether a multimodal music-understanding system's graph branch depends on its intended topology, and found that the answer needed to be broken into several smaller questions before it could be answered at all. An initial graph-randomization intervention turned out not to intervene, because the underlying construction was too dense. Once repaired and verified, we found no statistically reliable training-time advantage from the intended topology across nine dataset, neighborhood-size, and pooling combinations. "Undetected" is the honest word throughout, not "absent," since several intervals remain wide enough to permit practically meaningful effects. A trained model can nonetheless become sensitive, at test time, to topology perturbations in one specific regime: three of eighteen contrasts, spanning both destruction conditions, survive correction for multiple comparisons, all in GTZAN $k=4$. A post-hoc check confirms the three regimes are not all equal, and that this regime reliably exceeds one of the other two, though not the third. Removing edges entirely causes a large, reliable drop in performance, showing that the model depends on neighbor aggregation being present. At training time, however, we did not detect a reliable dependence on which particular neighbors are

aggregated, in any of the nine regime/pooling combinations tested, GTZAN $k=4$ included; the one exception above is specific to test time. In the deployed fusion system, pairing a caption with a different track's graph costs no statistically reliable disadvantage, under a fixed mismatch mapping or across ten independently sampled mismatch mappings evaluated hierarchically over four independently-trained checkpoints. This persists after removing literal tag-vocabulary matches and applying a matched token-count control, a narrower claim than saying correspondence goes completely unused.

None of this supports a claim that graphs are unhelpful for music understanding in general, that this system's graph branch could never be useful with a different architecture, or that the intended topology is definitively unused. It only shows that we did not detect reliable use of it here, on the audit construction rather than the original deployed graph directly. What it does support is narrower, and we think still useful: structural presence, edge presence, topology identity, and cross-modal correspondence are four different questions. Answering only the first one – does the pipeline have a graph branch – says little about the other three.

The density defect that started this audit is not specific to this system. Any similarity-threshold construction over a compact feature space – a common pattern whenever a graph is built directly from the same features that describe its nodes – can produce a near-complete graph without anyone noticing, silently disabling the randomization controls meant to test it. A paper reporting "we randomized the graph and performance did not change" is not evidence of topology-independence if the randomized graph was, in practice, almost the same graph. We hope the checklist in Section 7 helps practitioners check for this, and ask the right one of the four questions above, at the strength their evidence supports, before crediting or dismissing a feature-induced graph's contribution.

A.1 THE NON-FEATURE-INDUCED EDGES (TEMPORAL ADJACENCY)

Section 3's proposition applies to the similarity edges A_{sim} specifically, not the temporal-adjacency edges A_{temp} , since segment order is not recoverable from the unordered feature multiset. Auditing this against the source code found two verified implementation gaps, not information-theoretic impossibilities: the encoder never reads per-edge attributes (temporal and similarity edges are indistinguishable connectivity), and temporal edges are added symmetrically, so a segment sequence and its time-reversal are graph-isomorphic under this construction. Edge-conditioned message passing (Gilmer et al., 2017) and direction-aware GraphSAGE (Rossi et al., 2024) are off-the-shelf fixes; Section 6.5's pilot fixes the second

gap and tests the resulting signal, which does not help within that pilot's power.

A.2 REPRODUCIBILITY

Most statistics in Sections 5–6 were computed by a single analysis script (SHA256 9506ad476901a2c0, first 16 hex characters) reading only frozen raw result files, across an initial $n=3$ freeze and a later, approved extension to $n=8$ seeds plus two further checks; superseded estimates are marked as such in the text. The reference-construction density audit and cross-track measures use their own frozen output files instead. The script, output, seeds, splits, and software environment are archived in the supplement, whose script hashes serve as the reproducibility anchor in place of a version-control commit identifier.

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